Progress Report 3

1. **Project title:**
   Swarm of robots to complete locomotive and object-searching tasks, and touchscreen interface for a multipurpose robot.

2. **Names, student numbers, and emails**:
   Group 36 members: (all members are registered in ECSE 458-D1)

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3. **Project advisor(s):**
   Joseph Vybihal: joseph.vybihal@mcgill.ca

4. **Group meetings and meetings with advisor(s):**
   All four team members, along with the advisor, have agreed to hold two meetings each week: one with all team members present and another with the advisor and any available team members at that time.

   Weekly meeting with the entire group: Tuesdays at 3:45pm (previously it was Thursdays)
   Weekly meeting with the advisor: Fridays at 2pm

The report **(maximum 2 pages)** should include the following sections:

1. **Recent progress** (1/2 page maximum)**:** Summarize your progress so far.

   Since the last progress report, we started the implementation of the app. We switched our previous idea of doing a website with an Android App instead as we thought it better suited our requirements. Setting up the project and figuring out the languages and scripts we will use took a bit of time. The frontend of the app was started using React Native. The backend that deals with the user input from the app is going to be straightforward with local function calls (in JavaScript) that manipulates the data the user sends and will be stored in a csv file.

   We also started implementation of the backend development for the communication between the ESP 32, the tablet, and the Telenet server. We began by debugging the previous group's code in order to get it to run on the ESP32. There were some tricky

issues to sort out there, but once that was fixed we could move on to making some positive additions. We are able to establish a serial communication with the ESP 32 and change the code so it updates a specific register that is associated with different functions being run on the ESP32. We can send move commands to the ESP32 and see confirmation that these commands were sent.

In addition, the Telnet server is now up and running, and move commands can be sent from a laptop over WiFi with confirmation through the serial output.

2. **Future plans** (1/2 page maximum)**:** Summarize your future work plans for the period until the next report is due and provide an expected timeline for your progress.

Our future plan includes mainly finishing the code for the app. Followed by launching the app with the android tablet itself and making sure everything works as intended, so that we are on track with our initial goals of finishing the app in this semester. More specifically, success for our team will be establishing a working front end on an android tablet that can demonstrate a connection to the ESP 32 which in turn is connected to the Telnet server. Both ends of this connection chain should be able to read and write specific data to the opposing device. We will also be focusing on working on the midterm progress report that is due on the 18th December. Over the break, we plan on tying all loose ends and wrapping up the app and then spending some time to get familiarised with the swarm robots and get up to date with its documentation, such that we will be set up to start working on them next semester.

3. **Ethics and Equity** (1 page maximum)**:** Did you encounter any ethical dilemmas during this project or dilemmas related to equity? If so, how did you deal with them? If not, identify some possible issues that one may encounter in engineering projects in general.

Throughout the progress of our project, we did not encounter any explicit ethical dilemmas or concerns relating to equity. However, a number of potential ethical and equity-related problems may arise in projects like ours, especially in developing systems for public use in sensitive environments like hospitals. Below, we have outlined some of these possible challenges that we may encounter in the future and discuss strategies to mitigate them.

- Reliability of the system in the critical enviroment: Due to the fact that our robot's primary task is to perform tasks in hospital settings, such as delivering medications. System failures might delay important deliveries, and that could

result in accidents that would jeopardize patient care and safety. This raises ethical concerns about the reliability and robustness of the system.
Mitigation strategy/ Solution: In order to address this issue, we plan to focus on thorough testing throughout the development process to ensure everything runs smoothly. A big part of this will be making sure the Android tablet, ESP32, and Telnet server communicate reliably by testing the software at every step. Moreover, we plan to implement fail-safe mechanisms within the software architecture and add backup communication methods to the system, to ensure that even if something goes wrong, the robot can still function without major disruptions.

- Privacy and Security of Data: The touchscreen interface on the robot may handle sensitive task-related data, such as delivery schedules or individual patient-related personal information for the assigned task. Unauthorized access or data breaches could compromise the privacy of individuals in the hospital environment.

  Mitigation Strategy/ Solution: To address this, we will keep all data stored locally (e.g., in CSV files) and avoid using external servers or cloud platforms for data storage, this may reduce the chances of data breaches. If we stay on schedule, we can also plan to implement basic security measures, such as user authentication, to ensure that only authorized personnel can access and interact with the system.

- Accessibility and Usability: The robot's touchscreen interface must be accessible to a diverse range of users, including those with different levels of technical skills or physical limitations. Designing an interface that excludes certain users could lead to inequity.

  Mitigation Strategy/ Solution: We plan to design a simple and intuitive interface with clear visual cues to make it user-friendly for everyone. Features like visual feedback—such as the robot displaying expressive faces to show its status—may also help users understand its actions more easily. The interface will also include large, clearly labeled buttons for key tasks, which will ensure that users can easily perform actions like scheduling deliveries or confirming pickups without confusion or much complications.